



D. Y. Patil College of Engineering & Technology, Kolhapur

Assignment No. 4

Title: Implementation of Different layouts in

Android.

Theory:

-i- Introduction:

In Android, layouts are used to define the structure and appearance of the user interface. A layout controls how UI components such as button, text field, and image are arranged on the screen.

Proper use of layouts helps create responsive and user-friendly applications.

-ii- Information about layouts.

An Android Layout is a ViewGroup that arranges child views in a specific manner.

Layouts are defined using XML files, which separate the UI design from application logic. Layouts help in organizing UI components efficiently and adapting the interface to different screen sizes and orientations.

-iii- Commonly used layout attributes:

Some commonly used attributes in Android

layouts are:



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① android:layout_width - Specifies the width of a view

② android:layout_height - Specifies the height of a view

③ android:id - uniquely identifies a view.

④ android:layout_margin - Adds space outside the view

⑤ android:padding - Adds space inside the view

⑥ android:layout_gravity - Aligns the view within its parent

⑦ android:gravity - Aligns content inside the view.

⑧ android:orientation - Specifies horizontal or vertical arrangement (used in Linear Layout)

- Information about Android Layouts.

1. Linear Layout

Linear layout arranges child views in a single direction, either vertically or horizontally

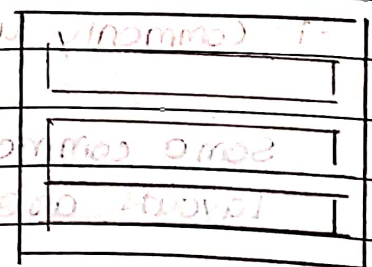


Horizontal

① Use android:orientation attribute

② Supports layout_weight to divide screen space proportionally

③ Best suited for simple UI designs.



Vertical



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2. relative Layout.

Relative layout positions views relative to the parent layout or other views.

It provides greater flexibility compared to LinearLayout

- Views can be aligned above, below, left or right of other views
- Reduces the need for nested layouts
- Improve UI performance

3. Constraint Layout.

Constraint Layout allows views to be positioned using constraints that define relationships between views.

- Recommended by Google for modern apps.
- Helps build complex layouts with a flat hierarchy.
- Improve performance and UI responsiveness.



4. Frame Layout.

FrameLayout is designed to hold a single view, but multiple views can be stacked on top of each other.

login Details
Enter your email
Enter password
(submit)

① last added view appears on top

② Commonly used for fragments

③ simple and lightweight layout

5. Table Layout.

TableLayout organizes views in rows and columns similar to an HTML table.

Row 1		
Row 2 Col 1	Row 2 Col 2	
Row 3 Col 1	Row 3 Col 2	Row 3 Col 3
Row 4		

① uses TableRow for creating rows

② columns automatically adjust based on content

③ useful for forms and structured data.



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6. Grid Layout

Grid layout arranges views in a rectangular grid of rows and columns.

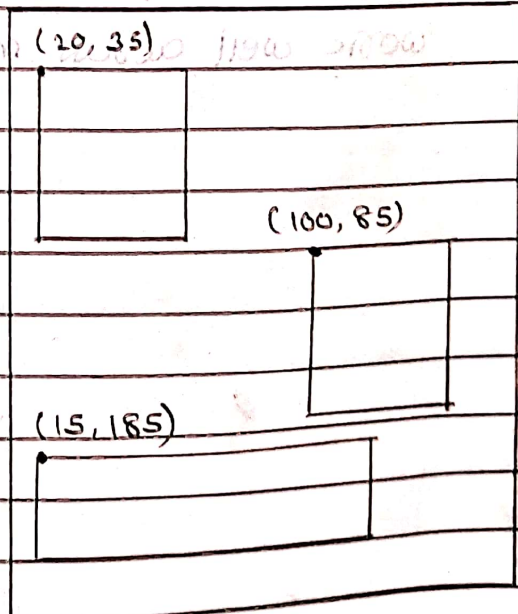
- More flexible than the Table Layout
- Allows views to span multiple rows or columns
- Suitable for dashboards and image grids.

item1	2	3
4	5	6
7	8	9

7. Absolute Layout (Deprecated)

Absolute layout places views using fixed x and y coordinates.

- Not responsive to different screen sizes
- Deprecated and not recommended
- Causes layout issues on multiple devices.





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8. ListView.

ListView displays a vertically scrollable list of items.

List 1

List 2

List 3

① Use adapters to bind data

① Efficient for displaying large datasets

① Supports scrolling and item click handling.

Conclusion:

Android layouts are essential for building responsive and visually appealing user interfaces. Understanding different layouts and their properties enable developers to design efficient applications that work well across multiple devices and screen sizes.